PRODUCT USER MANUAL

For
OMI_HEALTH_TEMPSALOXY_BALTIC_mbi_sto2tz_gotland
Issue: 4.0

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RECORD TABLE

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GLOSSARY AND ABBREVIATIONS

MOI	Mercator Ocean International
NetCDF	Network Common Data Form
OMI	Ocean Monitoring Indicator
PUM	Product User Manual
QUID	Quality Information Document
MBI	Major Baltic Inflow

DATA ACCESS

After registration, you will be able to download our data. To assist you, our HelpCenter is available, and more specifically its section about download.

Information on operational issues on products and services can be found on our User Notification Service. If needed, you can contact us .

1) INTRODUCTION

a) Summary

This document is the user manual for the Copernicus Marine Service Ocean Monitoring Indicator (OMI) OMI_HEALTH_TEMPSALOXY_BALTIC_mbi_sto2tz_gotland. It contains one dataset:

- omi_health_tempsaloxy_baltic_mbi_sto2tz_gotland

The OMI provides set of parameters which describe Major Baltic Inflows (MBI) in the southern and central Baltic Sea. The vertical profiles at the central and deepest Gotland Basin show the downstream reach of the MBI water. The oxygen profiles are indicator for the oxygen conditions in the deep central Baltic Sea, where anoxic bottoms prevail between the periods of the MBIs. The bottom salinity is found based on Baltic Sea insitu product INSITU_BAL_PHYBGCWAV_DISCRETE_MYNRT_013_032.

The product is published on the Copernicus Marine Service dissemination server after automatic and delayed-mode quality controls. The product is available online and disseminated through the Copernicus Marine Service Information System. Files downloaded are in NetCDF format and follow CF-1.7 convention.

Disclaimer: The quality of the product may vary during the proposed time series depending on the possible update of the system.

The OMI is further described in the corresponding Quality Information Document (QUID).

2) DESCRIPTION OF THE PRODUCT SPECIFICATION

a) General Information

Product Lines	OMI_HEALTH_TEMPSALOXY_BALTIC_mbi_sto2tz_gotland
Geographical coverage	Baltic Sea
Variables	Salinity, temperature, oxygen
Available time series	From 1993 to Month-4
Temporal resolution	Monthly
Horizontal resolution	One station
Integration depth (for example)	0 – 250 m
Format	NetCDF4

b) Production System Description

The current Major Baltic Sea OMI consist of vertical profiles of water salinity, temperature and oxygen concentration in the Gotland Deep area of the central Baltic Sea. In principle those are the key parameters that show the signal of inflowing water mass to the Baltic Sea regardless of data source.

The vertical profiles from Gotland Deep have been extracted from Copernicus Marine Service observations product for the Baltic Sea [INSITU_BAL_PHYBGCWAV_DISCRETE_MYNRT_013_032]. Station BY15, that is located at Gotland Deep, is part of the Swedish marine monitoring programme, covering the Skagerrak, the Kattegat, the Sound and the Baltic Proper. With a sampling frequency of once per month the profiles of temperature [°C], salinity [0.001] and dissolved oxygen [ml/l] give an overview of the deep-water processes at BY15, with special focus at the change between oxic/anoxic periods. All available observations from Copernicus Marine Service database for the Gotland Deep have been monthly averaged and interpolated on a regular depth grid. The months with missing data have been interpolated from subsequent months. The deepest value of each measurement are shown as a grey dot in the bottom of the figure.

c) Dataset details

omi_health_tempsaloxy_baltic_mbi_sto2tz_gotland contains the following variables:
so [0.001] long_name: Salinity profiles at Gotland deep standard_name: sea_water_salinity **to** [degree_C] long_name: Temperature profiles at Gotland deep standard_name: sea_water_temperature **oxy** [0.001] long_name: Oxygen profiles at Gotland deep standard_name: volume_fraction_of_oxygen_in_sea_water **time** [days since 1950-01-01 00:00:00] long_name: time standard_name: time **depth** [m] standard_name: depth

3) FILE FORMAT

The products are stored using the NetCDF format.

To know more about the NetCDF format, please follow this link:
What is the format of Copernicus Marine products ? NetCDF

To understand the differences between netCDF and Zarr, please consult this article:
how-to-choose-between-netcdf-and-zarr-format-using-the-toolbox

4) FILES NOMENCLATURE

Information about nomenclature of files when downloaded can be found in this article: " How is defined the nomenclature of Copernicus Marine data? | Copernicus Marine Help Center "

Files nomenclature when using the get command in the Copernicus Marine Toolbox or using the Browse File is as follows:

<datasetname>_P<productiondate>.nc

Where :
<datasetname> is the dataset reference
<productiondate> is the production date when the file was generated
.nc is the standard NetCDF filename extension.

5) REFERENCES

Quality Information Document (CMEMS-BAL-OMI-QUID-HEALTH-TEMPSALOXY-mbi-sto2tz-gotland)